Set
Mon 14 Sep 2026

Problem Set 6 — Mathematical induction

Session 6 · Ordinary induction, strong induction, sums, divisibility, and common errors

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

NOTATION $d \mid N$ means that $N = dq$ for some integer q. In words, "d divides N" means that N is a multiple of d.

FOR EACH INDUCTION PROOF, MAKE THESE PARTS EASY TO FIND

1. the statement $P(n)$ and its starting value;
2. the base case;
3. the inductive hypothesis for an arbitrary allowed integer k;
4. the inductive step from $P(k)$ to $P(k+1)$;
5. the conclusion by induction.

Part A · Start with the structure

1 NAME THE THREE MAIN PARTS

For the claim

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \qquad (n \geq 1),$$

write:

1. the base-case statement $P(1)$;
2. the inductive hypothesis $P(k)$;
3. the exact statement $P(k+1)$ that you must prove.

Do not prove the claim in this question. Write the three statements clearly.

2 SUM OF ODD NUMBERS

Prove by induction that, for every integer $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

In your inductive step, explain why the next term is $2k+1$.

3 AN EXPONENTIAL INEQUALITY

Prove by induction that, for every integer $n \geq 0$,

$$2^n \geq n + 1.$$

In the inductive step, state why $2(k+1) \geq k+2$.

Part B · Write complete proofs

4 DIVISIBILITY WITH AN EXPONENTIAL

Prove by induction that, for every integer $n \geq 1$,

$$5 \mid (6^n - 1).$$

Write the inductive hypothesis in the form $6^k - 1 = 5m$ for some integer m. In the next case, end with 5 times an integer.

5 SUM OF SQUARES

Prove by induction that, for every integer $n \geq 1$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

In the inductive step, factor out $k+1$ before simplifying.

6 A POLYNOMIAL IS ALWAYS DIVISIBLE BY 3

Prove by induction that, for every integer $n \geq 1$,

$$3 \mid (n^3 - n).$$

Write the inductive hypothesis in the form $k^3 - k = 3m$ for some integer m.

7 STRONG INDUCTION — POSTAGE USING 4 AND 5

Prove by **strong induction** that every integer $n \geq 12$ can be written as

$$n = 4a + 5b$$

for non-negative integers a and b. Here, non-negative means $0, 1, 2, \ldots$.

1. Show the four base cases $n = 12, 13, 14, 15$.
2. For the step, let $k \geq 15$ and assume that every integer from 12 to k has this form.
3. Use the earlier number $k - 3$ to prove the result for $k + 1$.

State why the strong hypothesis includes $k - 3$.

Part C · Read a proof critically

8 THE "ALL HORSES HAVE THE SAME COLOUR" ERROR

Someone tries to prove the false claim below by induction.

> **THE CLAIM** Claim $P(n)$: Every group of n horses has horses of only one colour.
>
> **Base case:** A group of one horse has only one colour.
>
> **Inductive step:** Assume $P(k)$. Take a group of $k+1$ horses. The first k horses have one colour by $P(k)$. The last k horses also have one colour by $P(k)$. These two groups share $k-1$ horses, so all $k+1$ horses have the same colour.

Answer all three questions.

1. Which value of k makes the inductive step fail?
2. Why does the "shared horses" sentence fail at this value?
3. Explain why adding more base cases cannot prove the original claim.

> **SELF-CHECK BEFORE SUBMITTING**
>
> 1. Did I write the correct starting value for every proof?
> 2. Did I state an inductive hypothesis for one allowed integer k?
> 3. Did I use that hypothesis in the step?
> 4. Did I prove the exact next statement, $P(k+1)$?
> 5. Did I write a conclusion that gives the correct range of n?